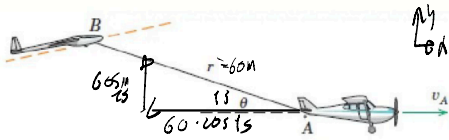


3. Airplane A is flying horizontally with a constant speed of 200 km/h and is towing the glider B, which is gaining altitude. If the tow cable has a length $r=60$ m and θ is increasing at the constant rate of 5 degrees per second, determine the magnitudes of the velocity and acceleration of the glider for the instant when $\theta=15^\circ$.



$$200 \frac{\text{km}}{\text{h}} \cdot \frac{1000 \text{ m}}{\text{km}} \cdot \frac{1 \text{ h}}{3600}$$

$$= 55.55 \text{ m/s}$$

$$\dot{\theta} = 5 \text{ deg/s} \rightarrow \frac{5 \text{ deg}}{s} \cdot \frac{\pi \text{ rad}}{180^\circ} = 0.08727 \text{ rad/s}$$

$$V_B = V_A + \omega \times r$$

$$V_B = 55.55 \hat{i} + \begin{vmatrix} 0 & 0 & -0.08727 \\ -57.95 & 15.52 & 0 \end{vmatrix}$$

$$V_B = 55.55 \hat{i} + 1.354 \hat{i} + 5.057 \hat{j} + 0 \hat{k}$$

$$V_B = 56.90 \hat{i} + 5.057 \hat{j} + 0 \hat{k} \text{ m/s}$$

Acceleración

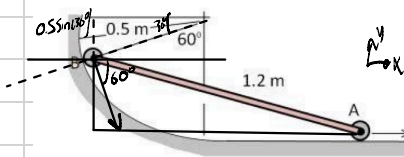
$$A_B = A_A + \alpha \times r + \omega(\omega \times r)$$

$$\dot{\theta} = 5 \text{ deg/s} \Rightarrow \ddot{\theta} = 0 = \alpha$$

$$A_B = \begin{vmatrix} 0 & 0 & -0.08727 \\ 1.354 & 5.057 & 0 \end{vmatrix}$$

$$A_B = 0.44 \hat{i} - 0.118 \hat{j} + 0 \hat{k} \text{ m/s}^2$$

At the instant represented, the velocity of point A of the 1.2-m bar is 3 m/s to the right. Determine the speed v_B of point B and the angular velocity ω of the bar. The diameter of the small end wheels may be neglected.



$$V_B = V_A + \omega \times r$$

Resolviendo el sistema

$$\frac{3 + 0.25\omega_K}{\cos 60} = -1.35\omega_K$$

$$\omega_K = -3.24 \text{ Rad/s}$$

$$\Rightarrow V_B = -1.35 \cdot -3.24$$

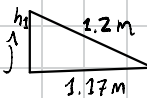
$$V_B = 4.38 \text{ m/s}$$

$$\omega \times r = \begin{vmatrix} i & j & k \\ \omega_i & \omega_j & \omega_k \\ r_i & r_j & r_k \end{vmatrix}$$

$$= \omega_j r_k - \omega_k r_j \hat{i} - (\omega_i r_k - \omega_k r_i) \hat{j} + (\omega_i r_j - \omega_j r_i) \hat{k}$$

$$V_{Bx} = V_B \sin 60^\circ$$

$$V_{By} = V_B \cos 60^\circ$$



$$0.5 - 0.5 \sin 30 = h_B$$

$$h_B = 0.25 \text{ m}$$

Para x

$$V_B \cos 60 = 3 + (\omega_K r_K - \omega_K r_j)$$

$$V_B \cos 60 = 3 - (\omega_K \cdot 0.25)$$

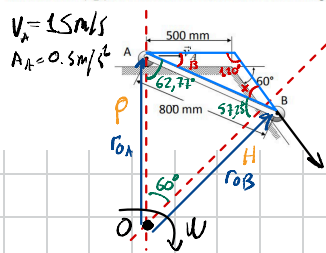
$$V_B = \frac{3 + 0.25\omega_K}{\cos 60} \quad (1)$$

Para y

$$-V_B \sin 60 = \omega_K r_i \Rightarrow V_B = -1.35\omega_K \quad (2)$$

7. Small wheels attached to both ends of rod AB roll along smooth surfaces.

At the instant shown, the wheel A has a velocity of 1.5 m/s towards right and its speed is increasing at a rate of 0.5 m/s^2 . Determine the absolute velocity of wheel B ($\vec{v}_B$), its relative velocity with respect to A ($\vec{v}_{B/A}$), its absolute acceleration ($\vec{a}_B$) and its relative acceleration with respect to A ($\vec{a}_{B/A}$).



$$V_A = V_A + V_{B/A} \quad V_{B/A} = \omega \times r$$

$$q_A = A_A + \alpha \times r + \omega(\omega \times r)$$

$$\frac{0.8}{\sin 120} = \frac{0.5}{\sin 60}$$

$$\alpha = 32.77^\circ$$

$$\beta = 27.23^\circ$$

$$p = \frac{800}{\sin 60} = \frac{p}{\sin 57.23}$$

$$p = 0.77 \text{ m}$$

$$H = \frac{800}{\sin 60} = \frac{H}{\sin 62.77}$$

$$H = 0.82 \text{ m}$$

$$V_{insA} = \omega \cdot r_{OA} \quad V_{insB} = 1.94 \cdot 0.82$$

$$1.5 = \omega \cdot 0.77$$

$$V_{insB} = 1.59 \text{ m/s}$$

$$\omega = 1.94 \text{ Rad/s}$$

$$V_B = V_A + V_{B/A}$$

$$V_B = 1.5 + V_{B/A}$$

$$V_{B/A} = 0.09 \text{ m/s}$$

$$q_B = q_A + \alpha \times r + A_B \hat{i}$$

$$\omega \times r = \begin{vmatrix} 0 & 0 & 1.94 \\ 0.71 & -0.34 & 0 \end{vmatrix}$$

$$= 0.65 \hat{i} + 1.137 \hat{j} + 0 \hat{k}$$

$$\omega \times (\omega \times r) = -2.65 \hat{i} + 1.26 \hat{j} + 0 \hat{k}$$

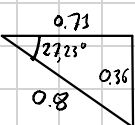
$$q_B \cos 60 = 0.5 + 0.36 \alpha_K - 2.65$$

$$-q_B \sin 60 = \alpha_K \cdot 0.71 + 1.26$$

$$0.5 q_B - 0.36 \alpha_K = -2.15$$

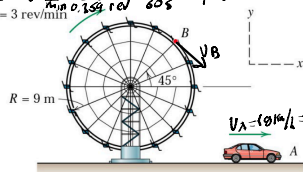
$$-0.86 q_B - 0.71 \alpha_K = 1.26$$

$$a_B = -2.97 \text{ m/s}^2 \quad \alpha = 1.86 \text{ rad/s}^2$$



1. The car A has a forward speed of 18 km/h and is accelerating at 3 m/s². Determine the velocity and acceleration of the car relative to observer B, who rides in a nonrotating chair on the Ferris wheel. The angular rate $\Omega = 3$ rev/min of the Ferris wheel is constant.

$$\omega = 3 \frac{\text{rev}}{\text{min}} \cdot \frac{1 \text{ rad}}{0.159 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 0.314 \text{ rad/s}$$



$$v_B = \omega \times r$$

$$v_B = \begin{vmatrix} 0 & 0 & -0.314 \\ 6.36 & 6.36 & 0 \end{vmatrix} = 1.99 \hat{i} - 1.99 \hat{j} + 0 \hat{k}$$

$$v_B = v_A + v_{B/A}$$

$$v_A = v_B + v_{A/B}$$

$$v_{B/A} = v_A - v_B$$

$$v_{B/A} = 5 \hat{i} - 1.99 \hat{i} + 1.99 \hat{j}$$

$$v_{B/A} = 3.00 \hat{i} + 1.99 \hat{j} \text{ m/s}$$

$$|v_{B/A}| = 3.606 \text{ m/s}$$

$$a_B = a_A + \omega \times r + \omega (\omega \times r)$$

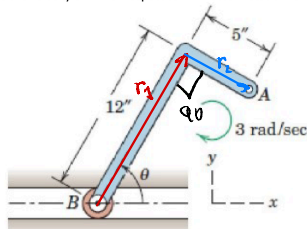
$$a_B = -0.62 \hat{i} - 0.62 \hat{j} + 0 \hat{k}$$

$$a_{A/B} = a_A - a_B$$

$$a_{A/B} = 3 - (-0.62 \hat{i} - 0.62 \hat{j} + 0 \hat{k})$$

$$a_{A/B} = 3.62 \hat{i} + 0.62 \hat{j} + 0 \hat{k}$$

The right-angle link AB has a clockwise angular velocity of 3 rad/sec at the instant when $\theta = 60^\circ$. Express the velocity of A with respect to B in vector notation for this instant.



$$r_1 = 12 \cos 60^\circ \hat{i} + 12 \sin 60^\circ \hat{j} + 0 \hat{k}$$

$$r_2 = 5 \cos (60 - 90)^\circ \hat{i} + 5 \sin (60 - 90)^\circ \hat{j}$$

$$r_{A/B} = 10.33 \hat{i} + 7.09 \hat{j}$$

$$v_{A/B} = \omega \times r_{AB} = \begin{vmatrix} 0 & 0 & -3 \\ 10.33 & 7.09 & 0 \end{vmatrix} = 23.67 \hat{i} - 30.99 \hat{j} + 0 \hat{k} \text{ in/s}$$